Comparison of Open Loop Optimal Control and Closed Loop Optimal Control of a Fermentation Process.

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Abstract

Optimisation of a fed-batch fermentation process is usually done by using the calculus of variations to determine an optimal feed rate profile. This often results in a singular control problem and an open loop control structure. To overcome these problems, the closed loop optimal control is developed by dividing the optimisation problem into two parts. First, an optimal substrate concentration profile which has direct effect to the biochemical reactions in the fermentation process is derived. Then a controller is designed to track the obtained optimal profile. A biomass production process is used to compare the performance of the closed loop optimal control method and the open loop optimal feed rate control method. The results show a better performance of the closed loop optimal control than the open loop optimal feed rate profile. This is due to the feedback property of the closed loop optimal control method.

Keyword : process control, optimisation, fermentation process

1. Introduction

Fermentation processes are used for producing many fine chemical substances such as amino acids, antibiotics, biomass, enzymes, etc. From modes of operation, (batch, fed-batch and continuous), fed-batch operation is often used in industry due to its ability to overcome the catabolite repression or glucose effect, which usually occur during production of these fine chemicals [1, 2]. Moreover, it also gives the operator the freedom of manipulating the process via substrate feed rate. This gives the challenge to the control and optimisation of the fed-batch fermentation processes.

Optimisation of fed-batch fermentation processes have been a topic of research for many years. To determine an optimal feed rate profile in the fed-batch fermentation, the other environment variables such as temperature and pH, which also affect bioreaction rates in the processes are assumed constant at some levels. The approaches used by many research groups to determine the substrate feed rate profile that optimises a desired objective function are usually based on the calculus of variations [3-8] or Green function [9]. And since there are physical constraints in the minimum and maximum feed rates, the Pontryagin's Maximum principle is applied. However, there are two problems arising in applying the variational method to the fermentation process. The first is that a singular control situation may occur during operation since the control input or the substrate feed rate appears linearly in the Hamiltonian. The other problem is that the obtained optimal feed rate profile from the variational method is in the open loop manner and can suffer quite severely when the model parameters are not exact when used in the real application. It is not until recently that a reliable neural network-based estimation of the substrate concentration has been developed and successfully implemented in industry [24] that we are ready to proposed a method that can avoid these problems [10]. The method separates the optimisation problem of the fermentation process into two parts. Firstly, the optimal substrate concentration profile which has direct effect to the biochemical reactions in the fermentation process is derived. Then a controller is designed to track the profile of the obtained optimal substrate concentration. With this approach, the singular problem is overcome, as the substrate concentration usually appears as a nonlinear function in the Hamiltonian. The open loop control is also converted into a close loop servo or regulator control. This method is then called here the "closed loop optimal control". The objective of this paper is to demonstrate the advantages of the closed loop optimal control method comparing to the open loop optimal feed rate control method.

The paper is structured as follows: Section 2 will describe the mathematical representation of a fed-batch fermentation process and formulate the closed loop as well as the open loop optimal control. The comparison between both methods is then shown in Section 3. The comparison is concluded in Section 4.

2. Closed Loop Optimal Control

The fed-batch fermentation can be represented by the following dynamic mass balance equations.

$$\frac{dX}{dt} = \mu X - DX \quad (1)$$

$$\frac{dS}{dt} = -\frac{1}{Y_{xs}}\mu X + D\left(S_f - S\right) \quad (2)$$

$$\frac{dP}{dt} = \pi X - DP \quad (3)$$

$$\frac{dV}{dt} = F \quad (4)$$

$$D = F/V \quad (5)$$

where X, S, P are biomass, substrate and product concentration (g/l) in the reactor respectively; F is the substrate feed rate (l/hr.); S_f is the concentration of substrate in the feed stream (g/l); D is dilution rate (1/hr.); μ and π are specific cell growth rate and specific product formation rate respectively (1/hr.); Y_{xs} is the yield of cell mass from substrate (g cell/g substrate) and V is fermenter volume (l). The specific rates μ and π are functions of substrate concentration. Further details and analysis on the fed-batch operation can be found in [2, 11, 12].

The fed-batch fermentation is constrained by conditions on final volume and minimum and maximum of substrate feed rates:

$$0 \leq F \leq F_{\max} \quad (6)$$

$$V\left(t_f\right) = V_f \quad (7)$$

The objective of the fermentation process is to produce as much product as possibly under production-time constraint. This objective is transformed into an objective function shown in Equation (8) and can be solved using the calculus of variations [13-15].

$$J(F) = f\left(X\left(t_f\right), P\left(t_f\right)\right) \quad (8)$$

The obtained open loop optimal feed rate profile consists of a sequence of maximum, minimum and singular feed rates depending on the following condition:

$$\frac{\partial H}{\partial F} = -\frac{\lambda_X X}{V} + \lambda_V + \frac{\lambda_S\left(S_f - S\right)}{V} - \frac{\lambda_P P}{V} = \Psi$$

From the Maximum principle, the optimal feed rate is determined by Ψ as follow:

if $\Psi < 0$ then $F = 0$

if $\Psi > 0$ then $F = F_{max}$

if $\Psi = 0$ then $F = F_{sing}$

The singular feed rate can be determined by repeatedly differentiating Ψ until feed rate (F) appears in the time derivative equation of Ψ.

$$\frac{d^k}{dt^k}\Psi = 0 \quad ; \quad k = 1, 2, 3, \ldots$$

The closed loop optimal control method separates the optimal control into two parts. The first part is to determine the optimal substrate concentration using the calculus of variations method but change the control variable from the substrate feed rate to substrate concentration in the fermenter. This eliminates the singular problem occurred in the open loop optimal feed rate profile method.

A controller can then be designed to track the obtained optimal substrate concentration profile and the control problem becomes a closed loop control problem. This also offers the flexibility that many types of controllers can be designed and used particularly those robust to model/process mismatch errors [16].

To determine the optimal substrate profile, the substrate feed rate and volume are omitted and the system equations become:

$$\frac{dX}{dt} = \mu X \quad (9)$$

$$\frac{dP}{dt} = \pi X \quad (10)$$

The objective function is then changed to:

$$J(S) = f\left(X\left(t_f\right), P\left(t_f\right)\right) \quad (11)$$

The optimal control can then be written down as:

$$\frac{\partial H}{\partial S} = \lambda_X X \frac{\partial \mu}{\partial S} + \lambda_P X \frac{\partial \pi}{\partial S} = 0 \quad (12)$$

And the optimal substrate concentration profile can be obtained by solving Equation (12) for the substrate concentration (S).

We use a nonlinear model predictive control scheme for the tracking control since the process model is available and nonlinear. More details on nonlinear model predictive control are in [17-23].

3.Comparison of open loop optimal feed rate profile (OLOFP) and closed loop optimal control (CLOC)

A biomass production process is used for demonstration and comparison between both methods. The objective of the process is to maximise the biomass production as shown in (13). The specific growth rate is assumed to be the substrate inhibition kinetic as shown in (14). The substrate inhibition kinetic is used in this study not only because it is simple to understand but also it can provide an analytical solution for comparison. Moreover, this type of kinetic can represent the catabolite repression and therefore the need for operating the fermentation in the fed-batch mode. Note that the model for this process consist of Equation (1), (2), (4) and (5).

$$J(F) = \max X\left(t_f\right) \tag{13}$$

$$\mu = \frac{\mu_{\max} S}{\left(K_S + S + S^2/K_i\right)} \tag{14}$$

It was shown in [10] that the optimal substrate concentration under the substrate inhibition kinetic can be calculated from the following condition, which is also the condition that the singular period occurs in the open loop optimal feed rate control:

$$\mu' = \frac{d\mu}{dS} = 0 \tag{15}$$

or

$$S_{opt} = \sqrt{K_S \cdot K_i}$$

For the open loop optimal feed rate profile method, the following feed rate can be obtained during the singular period:

$$F_{\text{sing}} = \frac{\mu X V}{Y_{xs}\left(S_f - S_{opt}\right)} \tag{16}$$

We consider here only the period in which the singular period occurs and therefore the substrate concentration is kept at the optimal from the beginning of the batch.

The following parameter values are used in the simulation:

$\mu_{max} = 0.10$, $K_s = 3$, $K_i = 8.34$ and $Y_{xs} = 0.164$
initial condition : X(0) = 1 g/l, S(0) = 5 g/l, and V(0) = 20 l
Final condition : $V(t_f) = 50$ l
substrate concentration in the feed stream:
$S_f = 100$ g/l
optimal substrate concentration:
$S_{opt} = \sqrt{K_S \cdot K_i} = 5$ g/l

In the closed loop optimal control, nonlinear model predictive control is used with the following parameters:

sampling time - 1 hr.
prediction horizon - 5 hr.
control horizon - 3 hr.

Since the CLOC method use the same process model as the CLOFP method to determine the optimal substrate concentration profile and feed rate, for an exact process model case, both methods give the similar results [10]. The comparison simulation here will therefore be performed under the condition of process-model mismatch where a small error (10%) is introduced to the parameter Y_{xs} to demonstrate this situation.

The simulation results for the OLOFP and CLOC methods are shown in Figure 1. It can be seen that in the modelling error case, the OLOFP method gives a wrong determination of the optimal feed rate. This results in the accumulation of substrate concentration in the fermenter which in turn reduces the growth rate and prolongs the operating time up to 85 hr. In the CLOC case, the substrate concentration is maintained at the optimal at 5 g/l for the whole batch until the reactor is full. The operating time for the OLOC method is only 72 hr.

The better performance of the CLOC than the OLOFP can be explained from the fact that the optimal singular feed rate needs accurate values of Y_{xs} and S_f to calculate the optimal feed rate as shown in (16), while the error in these two values can be compensated by feedback information in the CLOC method. Improvement of the CLOC method over the OLOFP method at different levels of error in the parameter Y_{xs} is shown in Figure 2.

Note that in both methods, the optimal and singular substrate concentration is determined by parameter K_s and K_i as shown in (15). Therefore, if these two parameters are inaccurate, both methods would result in the incorrect optimal substrate concentration profile and incorrect optimal feed rate profile.

However, the performance of the CLOC method is still better than the OLOFP method in these cases as shown from the shorter operating time in Table 1 where an error in parameter K_s is used for illustration. In this example, it is assumed that parameter K_s in the model equals 3, while this real parameter in the process equals 2 and 4. The incorrect optimal substrate concentration that is calculated from the model (K_s = 3) is 5 g/l while the true optimal levels are 4.083 g/l (K_s = 2) and 5.774 (K_s = 4).

It is shown in the table that for the process with K_s = 2, the operating time if the parameter in the model is correct is 68 hours. The operating time for the CLOC method is 68.5 hours, which is a little longer than the true one but shorter than that obtained from the OLOFP method, which is at 74 hours. For the process with K_s = 4, the operating time if the parameter in the model is correct is 81 hours. The operating time for the CLOC method is 82 hours and that obtained from the OLOFP method is 97 hours. The better performance of the CLOC method can be explained by the fact that keeping the substrate concentration at the incorrect optimal level (5 g/l) decreases the specific growth rate slightly from the optimal growth rate. While in the OLOFP method, the error in parameter K_s results in deviation of the optimal feed rate and substrate concentration, which in turn have highly effect on the biomass growth rate and prolong the process operating time.

This illustration shows the advantage of the CLOC method over the OLOFP method even in the case of incorrect parameter that is used in the optimal substrate concentration determination.

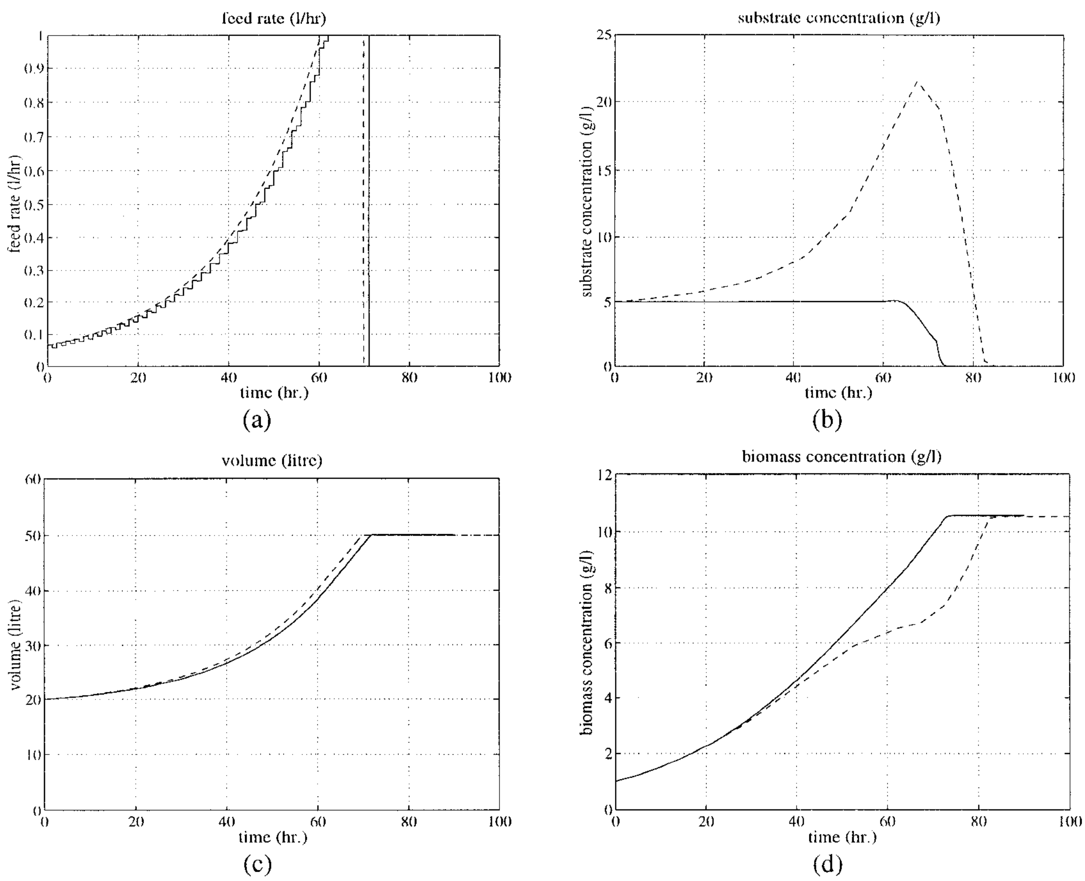

Figure 1 Comparison for both methods for plant/model mismatch in a biomass production process.

Solid line: closed loop optimal control method

Dashed line: open loop optimal feed rate profile method

% Shorter operating time of the closed loop over the open loop method compared with % error in parameter Yxs

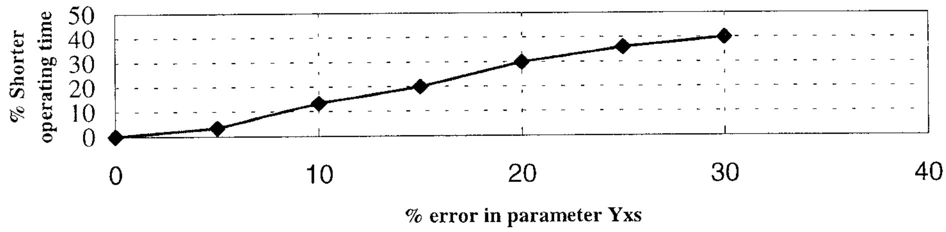

Figure 2 The closed loop method results in shorter operating time than the open loop method in the plant-model mismatch case.

Table 1 Effect of incorrect parameter K_s to the process operating time

Parameter K_s	Operating time (hr.)		
	Correct parameter for both methods	Incorrect parameter ($K_s = 3$)	
		CLOC	OLOFP
2 (-50%)	68	68.5	74
4 (25%)	81	82	97

4. Conclusion

The closed loop optimal control method divides an optimisation problem in a fed-batch fermentation into two parts.

1. determination of an optimal substrate concentration profile.
2. design a controller to follow the obtained optimal substrate concentration profile.

With this two steps procedure, the singular problem is overcome. And by converting the open loop into closed loop, it was shown by an example on a biomass production process that the closed loop optimal control provides a better results than the open loop optimal feed rate control method. This is due to the feedback property that provide the feedback information to compensate for the modelling error.

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